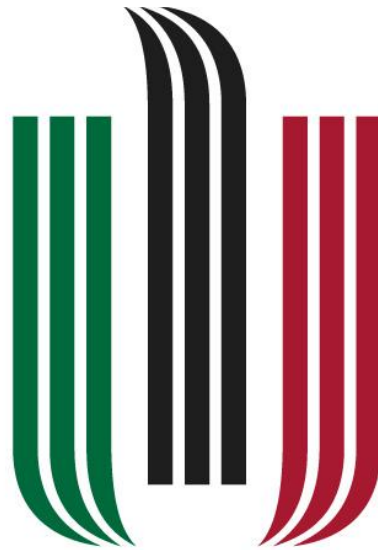




AGH

**AKADEMIA GÓRNICZO-HUTNICZA
IM. STANISŁAWA STASZICA W KRAKOWIE**

Advanced databases

Requirements for final project

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1. Project concept description

„Cryptocurrency market analysis system based on data from the CoinGecko API”

The aim of the project is to create a system for automatically collecting cryptocurrency market data, storing it in a database, and presenting the collected information in the form of interactive statistical visualizations.

The project will be based on market data related to selected cryptocurrencies, such as Bitcoin, Ethereum, Solana, or other chosen digital assets. The system will periodically retrieve data from the CoinGecko API and then store it in a database. The collected data will be used to analyze price changes, market capitalization, trading volume, and percentage changes in cryptocurrency values over time.

The main goal of the system is to enable users to observe market trends and compare the behavior of different cryptocurrencies within selected time intervals. The project will function as an analytical and statistical system designed for presenting and comparing market data.

2. Scope of analyzed data

The system will store and analyze, among others, the following data:

Data	Description
Cryptocurrency name	Full name of the asset, e.g. Bitcoin
Symbol	Cryptocurrency abbreviation, e.g. BTC
Price	Current price in a selected currency, e.g. USD, EUR, or PLN
Market capitalization	Total market value of a given cryptocurrency
Trading volume	Value of transactions in a given period
Percentage price change	Price change over 24 hours, 7 days, or another selected period
Market rank	Cryptocurrency position according to market capitalization
Measurement time	Date and time when the data was collected

3. Presentation of data sources

The data source used in the project will be the **CoinGecko API**.

CoinGecko provides a REST API in JSON format, which allows users to retrieve data related to the cryptocurrency market. According to the CoinGecko API documentation, the API provides access to data on cryptocurrency prices, market capitalization, trading volume, historical data, rankings, exchanges, and global cryptocurrency market data. CoinGecko states that it integrates data from over 1,000 exchanges and covers more than 18,000 cryptocurrencies across hundreds of categories.

In this project, endpoints that allow the retrieval of historical data will be particularly useful. The CoinGecko documentation describes, among others, the following endpoints: `/coins/{id}/market_chart`, `/coins/{id}/market_chart/range`, and `/coins/{id}/history`, which allow users to retrieve historical data related to price, market capitalization, and trading volume.

Example data retrieved from the API:

- current cryptocurrency price,
- historical price,
- market capitalization,
- trading volume from the last 24 hours,
- percentage price change,
- market capitalization ranking,
- cryptocurrency identifier,
- cryptocurrency symbol,
- date and time of measurement.

4. Example data requested from API:

Example below shows the data requested from **CoinGecko API**.

```
{
  "id": "bitcoin",
  "symbol": "btc",
  "name": "Bitcoin",
  "image": "https://coin-
images.coingecko.com/coins/images/1/large/bitcoin.png?1696501400",
  "current_price": 76182,
  "market_cap": 1525437558394,
  "market_cap_rank": 1,
  "fully_diluted_valuation": 1525438929779,
  "total_volume": 34007864832,
  "high_24h": 77432,
```

```
"low_24h": 75706,  
"price_change_24h": -619.7535500125232,  
"price_change_percentage_24h": -0.80695,  
"market_cap_change_24h": -12759531947.773926,  
"market_cap_change_percentage_24h": -0.82951,  
"circulating_supply": 20022003.0,  
"total_supply": 20022021.0,  
"max_supply": 21000000.0,  
"ath": 126080,  
"ath_change_percentage": -39.57641,  
"ath_date": "2025-10-06T18:57:42.558Z",  
"atl": 67.81,  
"atl_change_percentage": 112247.90356,  
"atl_date": "2013-07-06T00:00:00.000Z",  
"roi": null,  
"last_updated": "2026-04-28T17:46:16.705Z",  
"price_change_percentage_24h_in_currency": -0.8069524492506842,  
"price_change_percentage_7d_in_currency": 0.4987149980680044  
}
```